CRITICAL CARE MEDICINE · MIDTERM · LECTURE 3 · QUESTION BANK

Mechanical Ventilation

التنفس الصناعي — بنك الأسئلة

40	12	15
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

أسئلة سابقة · كويزات سابقة (طب حالات حرجة)

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 كويزات سابقة (طب حالات حرجة)

Mechanical ventilation is best described as:

a. Invasive or non-invasive
b. A permanent device
c. A new technology
d. Doesn't induce lung injury

2 كويزات سابقة (طب حالات حرجة)

Intrinsic PEEP, oxygen toxicity and pneumothorax are complications of mechanical ventilation.

a. True
b. False

3 كويزات سابقة (طب حالات حرجة)

A protective lung strategy in mechanical ventilation is to set TV at ..... to ..... ml/kg.

a. 1-3
b. 4-6
c. 7-10
d. 11-13

4 كويزات سابقة (طب حالات حرجة)

Which of the following are assist modes of ventilation? 1. Volume control 2. Pressure control 3. SIMV 4. CPAP

a. 1,3,4
b. 1,2,4
c. 2,3,4
d. 1,2,3

5 أسئلة سابقة

What is the recommended angle for the head of the bed in ventilated coma patients?

a. 0 degrees
b. 15 to 30 degrees
c. 30 to 45 degrees
d. 60 to 90 degrees

6 What is the purpose of a "Ventilator Bundle" in coma management? أسئلة سابقة

a. To prevent skin ulcers
b. To reduce the need for sedation
c. To prevent respiratory and circulatory complications
d. To ensure proper hydration

7 What does "ventilator bundle" refer to in the context of coma management? أسئلة سابقة

a. Measures for preventing respiratory and circulatory complications in ventilated patients
b. A treatment for electrolyte imbalance
c. A medication plan for coma patients
d. A nutritional intervention for coma patients

8 What is the primary goal of non-invasive ventilation in COPD patients during acute exacerbations? أسئلة سابقة

a. To prevent oxygenation failure
b. To improve blood sugar levels
c. To decrease sputum production
d. To increase lung volume

9 What is the goal of using non-invasive ventilation (NIV) in ARDS? أسئلة سابقة

a. To decrease blood pressure
b. To support lung function without intubation
c. To provide sedatives
d. To treat pneumonia

10 What is the most important component of ventilator care in ARDS patients? أسئلة سابقة

a. Use of high oxygen levels
b. Low tidal volumes and proper PEEP
c. Administration of antibiotics
d. Use of steroids for inflammation

11 What should be monitored closely in patients with ARDS who are on mechanical ventilation? أسئلة سابقة

a. Platelet count
b. Oxygenation and ventilation parameters
c. Blood glucose levels
d. Renal function only

12 What is the primary purpose of mechanical ventilation (MV)? أسئلة سابقة

a. To directly treat illnesses
b. To help a patient breathe when they cannot breathe on their own
c. To decrease blood pressure
d. To eliminate toxins from the blood

13 Which of the following is a non-invasive form of mechanical ventilation? أسئلة سابقة

a. Intubation
b. High Flow Nasal Cannula (HFNC)
c. Tracheostomy
d. Invasive Positive Pressure Ventilation (IPPV)

14 **Which condition is an indication for mechanical ventilation?** أسئلة سابقة

a. Mild Asthma exacerbation
b. Insufficient oxygenation (hypoxemia)
c. Controlled diabetes
d. Minor allergic reactions

15 **What is the primary benefit of mechanical ventilation in cases of insufficient ventilation?** أسئلة سابقة

a. It helps in removing excess oxygen
b. It aids in the removal of carbon dioxide (CO_2)
c. It helps reduce airway inflammation
d. It increases the amount of oxygen absorbed by the lungs

16 **In which of the following situations would mechanical ventilation be needed for airway protection?** أسئلة سابقة

a. Mild respiratory infection
b. Patient at risk of aspiration
c. Stable asthmatic patient
d. After a minor trauma

17 **Barotrauma is a complication of mechanical ventilation. What causes it?** أسئلة سابقة

a. Injury from excessive tidal volume and airway pressure
b. High oxygen levels causing tissue damage
c. Infection from the ventilator
d. Excessive CO_2 buildup in the lungs

18 **Auto-PEEP is a phenomenon in mechanical ventilation. What causes it?** أسئلة سابقة

a. Incomplete exhalation leading to positive pressure build-up
b. Excessive oxygen delivery during ventilation
c. Failure to initiate the first breath
d. Sudden reduction in ventilation rate

19 **Which of the following is not a type of mechanical ventilation?** أسئلة سابقة

a. High Flow Nasal Cannula (HFNC)
b. Helmet
c. Intubation
d. External Ventilation Mask

20 **What is the most common indication for mechanical ventilation in severe asthma cases?** أسئلة سابقة

a. Mild airway narrowing
b. Respiratory failure due to airway constriction
c. Chronic asthma management
d. Normal blood oxygen saturation levels

21 **What is the main purpose of using the Assist/Control (A/C) mode of ventilation?** أسئلة سابقة

a. To allow the patient to breathe spontaneously
b. To provide full ventilator support if the patient cannot breathe on their own
c. To help with the timing of spontaneous breaths
d. To assist the patient in weaning off mechanical support

22 In which mode of mechanical ventilation does the ventilator provide mandatory breaths at set intervals while allowing spontaneous breathing? أسئلة سابقة

a. Assist/Control (A/C)
b. Synchronous Intermittent Mandatory Ventilation (SIMV)
c. Pressure Support Ventilation (PSV)
d. Continuous Positive Airway Pressure (CPAP)

23 Pressure Support Ventilation (PSV) is most appropriate for which type of patient? أسئلة سابقة

a. A patient who is unable to breathe on their own
b. A patient who requires full ventilator support
c. A patient who can breathe independently but needs assistance with the effort
d. A patient with acute lung injury

24 Which of the following is true regarding Continuous Positive Airway Pressure (CPAP) mode? أسئلة سابقة

a. It provides complete control of breathing
b. It keeps the airways open without controlling the patient's breaths
c. It allows for spontaneous breathing only
d. It is used only during sleep

25 What is the main difference between Control Mode Ventilation (CMV) and other ventilation modes? أسئلة سابقة

a. The patient controls the timing of breaths
b. The ventilator provides all breaths with no spontaneous breathing allowed
c. The patient does not require any mechanical support
d. The mode alternates between high and low airway pressures

26 Which of the following is not included in the Ventilator Bundle for reducing complications like VAP? أسئلة سابقة

a. Head of bed elevation
b. Oral care with chlorhexidine
c. Daily sedation vacation
d. Increased oxygen saturation

27 What is the purpose of peptic ulcer prophylaxis in mechanically ventilated patients? أسئلة سابقة

a. To prevent respiratory infections
b. To prevent the development of stress ulcers
c. To promote lung expansion
d. To reduce inflammation in the lungs

28 Which of the following complications is most commonly associated with mechanical ventilation for extended periods? أسئلة سابقة

a. Barotrauma
b. Pneumothorax
c. Ventilator-Associated Pneumonia (VAP)
d. All of the above

29 High Flow Nasal Cannula (HFNC) provides what type of support in mechanical ventilation? أسئلة سابقة

a. High levels of oxygen delivery
b. Positive pressure ventilation
c. Complete mechanical ventilation support
d. Just oxygen delivery without pressure

30 In which situation is intubation the most appropriate intervention? أسئلة سابقة

a. A patient requiring short-term oxygen support
b. A patient with moderate asthma exacerbation
c. A patient with severe airway obstruction or respiratory failure
d. A patient needing oxygen at home

31 Which of the following is a benefit of mechanical ventilation? أسئلة سابقة

a. Decreases blood pressure
b. Increases patient activity levels
c. Facilitates carbon dioxide removal
d. Increases the production of red blood cells

32 Ventilator-Induced Lung Injury (VILI) can occur due to which of the following? أسئلة سابقة

a. High PEEP and excessive tidal volume
b. Low PEEP and low tidal volume
c. Low-frequency ventilation
d. Short-term oxygen therapy

33 The Synchronous Intermittent Mandatory Ventilation (SIMV) mode is most appropriate for patients who: أسئلة سابقة

a. Cannot breathe independently
b. Require full support for every breath
c. Are starting to breathe independently but need some ventilator assistance
d. Have complete lung collapse

34 In which of the following conditions would Mechanical Ventilation be contraindicated? أسئلة سابقة

a. Severe asthma exacerbation
b. Acute lung injury (ALI)
c. Cardiac arrest with no chance of recovery
d. Acute respiratory distress syndrome (ARDS)

35 Which ventilator mode is best suited for a patient who needs full ventilator support and cannot breathe on their own? أسئلة سابقة

a. Assist/Control (A/C)
b. Pressure Support Ventilation (PSV)
c. Synchronous Intermittent Mandatory Ventilation (SIMV)
d. Continuous Positive Airway Pressure (CPAP)

36 Which complication of mechanical ventilation results from prolonged exposure to high oxygen levels? أسئلة سابقة

a. Pneumothorax
b. Oxygen toxicity
c. Barotrauma
d. Hypercapnia

37 What is the goal of Daily Sedation Vacation in ventilated patients? أسئلة سابقة

a. To increase the level of sedation
b. To assess whether sedation can be reduced or stopped
c. To adjust the ventilator settings
d. To assess the patient's need for mechanical ventilation

38 High PEEP levels during mechanical ventilation may lead to which complication? أسئلة سابقة

a. Oxygen toxicity
b. Pneumothorax
c. Oxygen desaturation
d. Hypercapnia

39 Which of the following ventilator alarms alerts the healthcare team to a potential issue with the patient's ventilation rate? أسئلة سابقة

a. High Pressure alarm
b. High Frequency alarm
c. Low Pressure alarm
d. Apnea alarm

40 Mechanical Ventilation can be required for patients with severe hypotension due to which condition? أسئلة سابقة

a. Sepsis
b. Heart failure
c. Cardiogenic shock
d. All of the above

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 Indications for mechanical ventilation include (select all): SELECT ALL

a. Hypoxemic respiratory failure
b. Hypercapnic respiratory failure
c. Airway protection (GCS ≤ 8)
d. Cardiac arrest
e. Mild anxiety

N2 The high-pressure alarm on a ventilator may be caused by (select all): SELECT ALL

a. Secretions in the airway
b. Patient biting the tube or coughing
c. Kinked tubing
d. Disconnection of the circuit
e. Bronchospasm

N3 A ventilated patient suddenly desaturates. Using DOPE, the nurse checks for: NCLEX

a. Displacement, Obstruction, Pneumothorax, Equipment failure
b. Dehydration, Obesity, Pain, Edema
c. Drugs, Oxygen, Pressure, Electrolytes
d. Diet, Output, Position, Exercise

N4 **Elements of the VAP prevention bundle include (select all):** SELECT ALL

a. Head of bed 30–45°
b. Daily sedation interruption and readiness-to-wean assessment
c. Oral care with chlorhexidine
d. DVT and peptic ulcer prophylaxis
e. Routine daily circuit changes

N5 **Endotracheal cuff pressure should be maintained at:** NCLEX

a. 20–30 cmH_2O
b. 5–10 cmH_2O
c. 40–50 cmH_2O
d. 60 cmH_2O

N6 **During suctioning of an intubated patient, the nurse should (select all):** SELECT ALL

a. Pre-oxygenate with 100% O_2
b. Apply suction only while withdrawing
c. Limit each pass to 10–15 seconds
d. Use clean technique and reuse the catheter
e. Monitor SpO_2 and heart rate

N7 **PEEP improves oxygenation by:** NCLEX

a. Keeping alveoli open at end-expiration
b. Increasing respiratory rate
c. Reducing FiO_2 to zero
d. Increasing dead space

N8 **Which ventilator mode delivers a set tidal volume for every breath, whether triggered by the patient or the machine?** NCLEX

a. Assist/Control (A/C)
b. CPAP
c. Pressure support
d. SIMV

N9 **Readiness for weaning is suggested by (select all):** SELECT ALL

a. Underlying cause improving
b. Hemodynamic stability
c. $FiO_2 \le 40–50\%$ with PEEP ≤ 5–8
d. Adequate cough and mental status
e. High-dose vasopressors

N10 **Signs of failure during a spontaneous breathing trial include (select all):** SELECT ALL

a. RR > 35/min
b. $SpO_2 < 90\%$
c. Heart rate rising > 20% or arrhythmia
d. Diaphoresis and agitation
e. Calm, comfortable breathing

N11 **Correct ETT position is confirmed by (select all):** SELECT ALL

a. Waveform capnography ($ETCO_2$)
b. Bilateral equal breath sounds and chest rise
c. Chest X-ray
d. Patient talking normally
e. Absence of gastric sounds

N12 **Lung-protective ventilation in ARDS uses tidal volume of:** NCLEX

a. 4–6 (up to 8) mL/kg predicted body weight
b. 10–12 mL/kg actual weight
c. 15 mL/kg
d. 2 mL/kg

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Purpose of MV	Support oxygenation and ventilation (CO_2 removal), protect the airway, rest respiratory muscles.	**Invasive vs non-invasive**	Invasive: ETT / tracheostomy. Non-invasive: face mask, helmet (CPAP/BiPAP); HFNC gives oxygen.
Indications	Hypoxemia, hypercapnia/acidosis, apnea, GCS ≤ 8 / aspiration risk, shock, cardiac arrest.	**A/C**	Full support: set volume for every breath (patient- or machine-triggered).
SIMV	Mandatory breaths synchronized with spontaneous breaths — used in weaning.	**PSV**	Pressure boost for each spontaneous breath; patient sets rate.
CPAP	Continuous pressure, all breaths spontaneous; keeps airways open.	**CMV**	Machine controls all breaths; no spontaneous breathing.
Protective ventilation	TV 4–6 (≤ 8) mL/kg PBW, plateau ≤ 30 cmH_2O, adequate PEEP.	**Complications**	Barotrauma/pneumothorax, volutrauma, auto-PEEP, oxygen toxicity, VILI, VAP, hypotension, stress ulcer, DVT.
VAP bundle	HOB 30–45°, sedation vacation, chlorhexidine oral care, PUD & DVT prophylaxis.	**Alarms**	High pressure: secretions, biting, kink, bronchospasm. Low pressure: disconnection, cuff leak.
DOPE	Displacement, Obstruction, Pneumothorax, Equipment failure.	**Cuff pressure**	20–30 cmH_2O.
Contraindication	Futile cases (no chance of recovery / DNR).		

Answer Key

الإجابات

1. a — Invasive or non-invasive
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3. b — 4-6
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4. c — 2,3,4
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). إجابة الطالب في صورة الكويز. المحاضرة القديمة بتعد الـ modes: A/C, SIMV, PSV, CPAP, CMV.

5. c — 30 to 45 degrees

6. c — To prevent respiratory and circulatory complications

7. a — Measures for preventing respiratory and circulatory complications in ventilated patients

8. a — To prevent oxygenation failure

9. b — To support lung function without intubation

10. b — Low tidal volumes and proper PEEP

11. b — Oxygenation and ventilation parameters

12. b — To help a patient breathe when they cannot breathe on their own

13. b — High Flow Nasal Cannula (HFNC)

14. b — Insufficient oxygenation (hypoxemia)

15. b — It aids in the removal of carbon dioxide (CO_2)

16. b — Patient at risk of aspiration

17. a — Injury from excessive tidal volume and airway pressure

18. a — Incomplete exhalation leading to positive pressure build-up

19. d — External Ventilation Mask

20. b — Respiratory failure due to airway constriction

21. b — To provide full ventilator support if the patient cannot breathe on their own

22. b — Synchronous Intermittent Mandatory Ventilation (SIMV)

23. c — A patient who can breathe independently but needs assistance with the effort

24. b — It keeps the airways open without controlling the patient's breaths

25. b — The ventilator provides all breaths with no spontaneous breathing allowed

26. d — Increased oxygen saturation

27. b — To prevent the development of stress ulcers

28. d — All of the above

29. a — High levels of oxygen delivery

30. c — A patient with severe airway obstruction or respiratory failure

31. c — Facilitates carbon dioxide removal

32. a — High PEEP and excessive tidal volume

33. c — Are starting to breathe independently but need some ventilator assistance

34. d — Acute respiratory distress syndrome (ARDS)
من أهم دواعي الـ ARDS وده غلط لأن الـ ARDS، الملف كاتب MV. الإجابة الصح c (cardiac arrest with no chance of recovery / DNR).

35. a — Assist/Control (A/C)

36. b — Oxygen toxicity

37. b — To assess whether sedation can be reduced or stopped

38. b — Pneumothorax

39. b — High Frequency alarm

40. d — All of the above

N1. a, b, c, d — Hypoxemic respiratory failure; Hypercapnic respiratory failure; Airway protection (GCS ≤ 8); Cardiac arrest
Inadequate oxygenation, ventilation or airway protection.

N2. a, b, c, e — Secretions in the airway; Patient biting the tube or coughing; Kinked tubing; Bronchospasm
Disconnection or cuff leak triggers the LOW-pressure alarm.

N3. a — Displacement, Obstruction, Pneumothorax, Equipment failure
If unsure, disconnect and bag with 100% O_2.

N4. a, b, c, d — Head of bed 30–45°; Daily sedation interruption and readiness-to-wean assessment; Oral care with chlorhexidine; DVT and peptic ulcer prophylaxis
Circuits are changed only when visibly soiled.

N5. a — 20–30 cmH_2O
Low → aspiration/leak; high → tracheal ischemia.

N6. a, b, c, e — Pre-oxygenate with 100% O_2; Apply suction only while withdrawing; Limit each pass to 10–15 seconds; Monitor SpO_2 and heart rate
Use sterile technique (or a closed in-line system).

N7. a — Keeping alveoli open at end-expiration
High PEEP can reduce venous return (hypotension) and cause barotrauma.

N8. a — Assist/Control (A/C)
Risk: hyperventilation if the patient breathes fast.

N9. a, b, c, d — Underlying cause improving; Hemodynamic stability; $FiO_2 \leq 40–50\%$ with PEEP ≤ 5–8; Adequate cough and mental status
Spontaneous breathing trial (T-piece or low PSV) for 30–120 minutes.

N10. a, b, c, d — RR > 35/min; $SpO_2 < 90\%$; Heart rate rising > 20% or arrhythmia; Diaphoresis and agitation
Stop the trial and return to previous settings.

N11. a, b, c, e — Waveform capnography ($ETCO_2$); Bilateral equal breath sounds and chest rise; Chest X-ray; Absence of gastric sounds
Note the tube mark at the lips/teeth.

N12. a — 4–6 (up to 8) mL/kg predicted body weight
Plateau pressure ≤ 30 cmH_2O.